

THE TRAINING DATA LESSON • LESSON 2 OF 6

What's in the Playlist?



How AI learns — and where bias comes from. An AI is a music app, its training data is the playlist, and a one-vibe playlist (all loud, all fast, all party) means one-vibe answers. Stands alone.

AT A GLANCE

TIME	MATERIALS	TECH NEEDED	PREP
45 min	Playlist Check worksheet, 2 pages, 1 per student — each fills their OWN sheet (typeable PDF copy included) · Slides (optional; script in the notes) · board · scissors optional	None — runs fully unplugged. Optional: a projector for the deck	Print worksheets. That's it.

OBJECTIVES

- Explain how an AI learns: **input** → **patterns** → **output** — prediction from data, not magic. (*Exit Ticket #1*)
- Name who a dataset serves well and who it leaves out; link that gap to real AI bias. (*Gap Report; Exit #2*)
- Discuss data gaps collaboratively: repeat a partner's idea, then add to it. (*Repeat, Then Add; Exit Ticket #3*)

STANDARDS ALIGNMENT

CSTA 2-IC-21

ISTE 1.1.d

CCSS ELA SL.6.1

Full text: Teacher Guide, p.3

DIFFERENTIATION

- **Support:** Gray sentence starters on the worksheet; landmark-card tip in the Teacher Guide, p.2.
- **Extension:** Design a 12-card playlist that serves every request on the sheet, then defend the tradeoffs.
- **ELL:** Fit marks are checked boxes, not sentences; invented cards may be in any language (that fixes a real gap).
- **No-device classrooms:** Built in — the lesson is paper, pairs, and one loud imaginary music app.

🎵 WARM-UP — THE ONE-PLAYLIST APP (7 MIN)

You are **AutoMix**, the school's new music app. Your playlist holds one kind of song: loud, fast party tracks. Take 3–4 requests and serve every one with total confidence:

"Something chill to study to?" → "Here's 'TURBO TURBO.' It is not chill. It's what I've got."

"A song for grandma's birthday dinner?" → "Got it. 'VOLUME WAR.' Grandma loves bass drops, probably."

"A slow song for the last dance?" → "'NEON STAMPEDE,' played at... the exact same speed."

Then ask: **"Is AutoMix bad at picking songs — or is something else broken?"** Fish for it: the app is doing its job perfectly. The *playlist* is the problem.

If nobody gets there: "Could AutoMix play a slow song if it had one?" Yes — the app works. So what's missing?

Bridge: "(If your class did the genie lesson: it granted what you SAID.) Today: where an AI's answers even come from. Everything an AI 'knows' comes from a playlist of examples — its training data. It can remix the playlist, but it can't serve what was never in it. Today you'll BE the app."

Transition: "So what IS in an AI's playlist, and what does it do with it? Three words on the board."

DIRECT INSTRUCTION — NO MAGIC IN THE MACHINE (8 MIN)

INPUT — the playlist. Training data: every example the AI studied before you met it. Nobody types "knowledge" in.

PATTERNS — the training. It gets good at ONE thing: guessing what usually goes with what. No understanding — patterns.

OUTPUT — the mix. It predicts what best matches those patterns. A prediction, not a lookup — confident and still off.

Say it plainly (board: INPUT → PATTERNS → OUTPUT, one word per box): "No magic in the machine. An AI doesn't KNOW things — it predicts things, from its playlist. So two rules run everything: a GAP in the playlist is a gap in the AI, and a LEAN in the playlist leans every answer. That lean has a name: bias."

Quick check: "A photo AI's playlist is almost all golden retrievers. A husky walks in — what's it going to call it?" (Golden retriever — its best guess. Gap in, gap out.)

If a student says "husky": "Has this AI ever seen one? It can only pick from the patterns it has."

Transition: "Hand out Playlist Check. Pair up. Read the gray box at the top together — it names your two jobs."

ACTIVITY — PLAYLIST CHECK (20 MIN)

1. **Setup (2 min):** Partner A is the **App**, Partner B the **Caller**; the 12 cards ARE the app's training data. Rules: serve only from the playlist, never say no, check ONE Fit box honestly — each on their OWN sheet.

If there's an odd number: a trio — two Callers, one App, swap twice.

Transition: "Caller, read #1 out loud. App — you can't say no."

2. **Serve the Caller (8 min):** Six requests (Priya, Nia, Tomás, Keisha, Malik, Jordan): App names the card; each checks their OWN Fit box (they may differ). Swap at #4; end with one live request from a neighboring pair.

If a pair can't agree on the Fit: each checks their own — disagreement is data. Save it for the debrief.

Transitions: (#3) "Swap jobs: new App, new Caller. On to #4." (#7) "Turn to the pair beside you. Make up ONE request for their app — they can't say no." (after #7) "Pens on page 2. Say who it let down."

3. **Playlist Gap Report (7 min):** Each student writes who the playlist serves well and who it leaves out. Rule, enforced out loud: *repeat your partner's idea, then add to it* — both go on the Repeat, Then Add lines.

If a pair writes "it needs more songs": ask WHO asked for something that wasn't there.

Transition: "Three minutes. Fix the playlist — invent the cards that are missing."

4. **Fix the Playlist (3 min):** Invent new cards that fix the biggest gaps — at least 2. Push past "add one slow song": which requests still have nothing?

If every new card is a slow song: "Tomás still has nothing. What does he need?"

Transition: "Pens down. Eyes up. Who did your playlist let down? Hands."

DEBRIEF — THE REAL PLAYLISTS (5 MIN)

2–3 pairs name one gap (1 min). Three real playlists, Teacher Guide p.2 (3 min). Big question (1 min):

"Nobody put bad songs in your playlist — so where did the bias come from?"

Answer: what got collected, and what didn't. "On purpose?" Usually not — teams grab the easiest data to get.

Transition: "Back to your own sheet. The bottom of page 2 is yours alone — five minutes."

EXIT TICKET (5 MIN)

Worksheet p.2, solo: **#1** finish **input** → **patterns** → **output** + what an AI does with its playlist; **#2** diagnose a lopsided homework-helper AI using "playlist"; **#3** "One gap your partner spotted, and the idea YOU added to it." Key: Teacher Guide p.2.

If a student writes "the AI learns": "Learns from what? Say where the patterns came from."